



Stable Matchings

Lecture 6 | Algorithms Design & Analysis

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Matching Problem



Definition 1. A *bipartite graph* is a graph whose vertex set is partitioned into two disjoint sets L, R such that each edge has one endpoint in L and the other endpoint in R . When we write a bipartite graph G as an ordered triple $G = (L, R, E)$, it means that L and R are the two vertex sets (called the *left set* and *right set*, respectively) and E is the edge set.

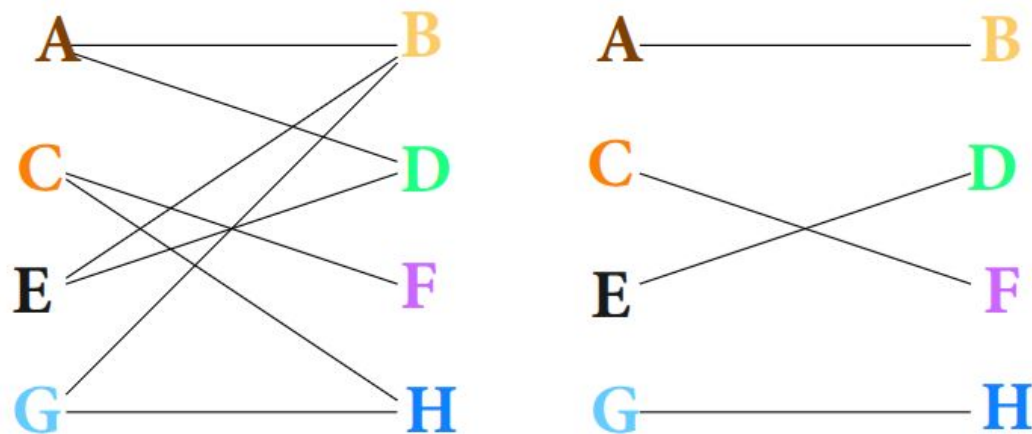
Definition 2. A *matching* in an undirected graph is a set of edges such that no vertex belongs to more than element of the set.

The *bipartite maximum matching problem* is the problem of computing a matching of maximum cardinality in a bipartite graph.

Matching Problem



Example: M is a perfect matching in G :

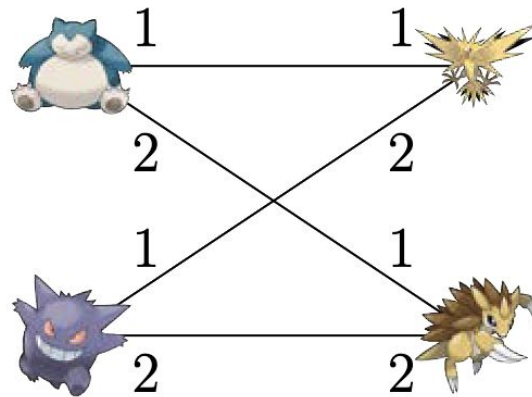


Stable Matchings



Bipartite Graph: N Men and N Woman

Each have a preference list

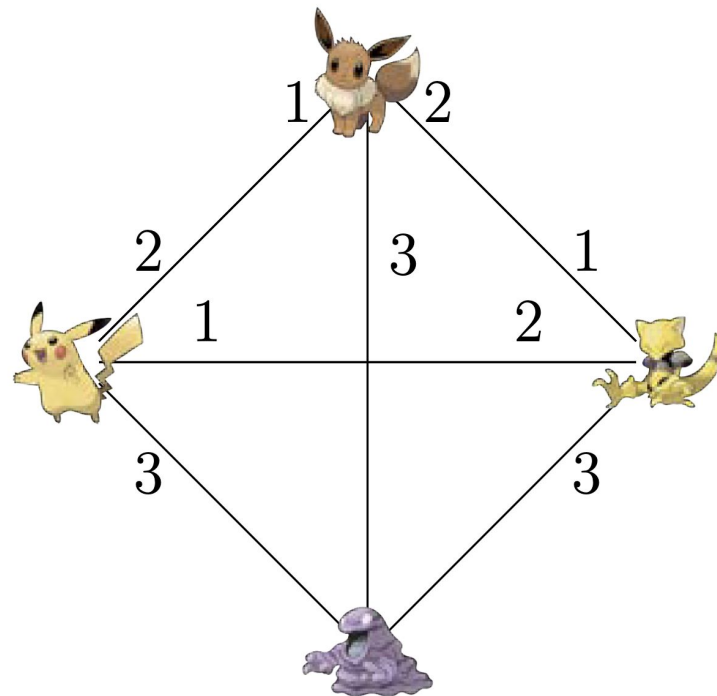


Definition 7. A rogue couple in a matching M is a pair (x, y) that are not matched in M but prefer each other over their matches.

Goal: Find a perfect matching without any rogue couples.

Example

Find a stable matching.



Gale-Shapley Algorithm



Repeat until all men are matched

1. Each woman proposes to most preferred man
2. Each man tentatively chooses the most preferred woman from whom he got proposals and rejects the others.
3. If a man who was tentatively matched previously, gets a better offer according to his preference list, he switches.
4. Woman who gets rejected, removes that man from their preference list.

Gale-Shapley Algorithm



Theorem 1: Algorithm terminates in N^2 steps.

At each step at least 1 man get removed from some women's preference list.

Theorem 2: Matches everyone.

Once a man is matched, he is never unmatched; he only swaps partners.

Gale-Shapley Algorithm



Theorem 3: produces a Stable Matching

Assume m and w form a blocking pair

Case #1: w never proposed to m

Case #2: w proposed to m

Optimal for Woman



Theorem: Algorithm results in every woman getting her best choice among all possible stable matchings.

Theorem: Algorithm results in every man getting his worst choice among all possible stable matchings.

Read



For Stable Matchings

- Watch:
 - <https://www.youtube.com/watch?v=Qcv1lqHWAzq>
 - <https://www.youtube.com/watch?v=LtTV6rlxhdo>
- Kleinberg Tardos, Chapter 1
<https://theswissbay.ch/pdf/Gentoomen%20Library/Algorithms/Algorithm%20Design%20-%20John%20Kleinberg%20-%20%C3%89va%20Tardos.pdf#page=13.40>